

ENCN442 Coastal Engineering Research Project

Due 10am, Monday 11th May

Project Description

You are required to produce a research report on a specific coastal engineering intervention. The target audience of your report is an engineering employer who is considering the intervention as part of a larger coastal project somewhere in New Zealand. Any coastal engineering intervention is fine, but some examples are a sea wall or beach nourishment project. The assignment involves six stages:

1. Using a large language model (e.g., ChatGPT or Claude), produce a report on your intervention. Include your prompt(s) and the response in your final submission.
 - As a guide, a response of around 750 – 1000 words should give sufficient detail for you to critique and still add additional detail.
 - Don't use extended research or multi-step agentic features (e.g, Deep Research, extended thinking modes). These will make your validation job more difficult.
 - Feel free to do some minor prompt engineering but don't go too far – again, an imperfect response will make your validation easier.
2. Use citable and reliable references to validate specific claims in your report.
 - It might not be feasible to validate all claims in a reasonable amount of time. I expect an average student would spend around 10 hours on this assignment, so let that be a guide.
 - It is fine to validate some claims and then note, by highlighting for example, other claims that should be validated.
3. Provide additional detail as needed where your LLM model output was lacking. I would hope that you can provide additional information beyond an LLM response. It's mainly up to you where you'd like to provide deeper layers of information and more detail. However, I do expect you to expand on bicultural considerations beyond what the LLM provides.
4. A reflection on the use of LLMs in engineering work. This could include best practices, ethical considerations, limitations and strengths, or anything else that you think is relevant and important. Relating your reflection to the topic of this assignment could help you frame it. This response should be around 300 words.
5. References to trusted sources that support specific claims that have been made. Trusted sources would be journal articles, design guides, technical specifications, etc.
6. An estimate of how long you've spent on the assignment. I expect the typical student to spend 10 hours. This won't be marked and is just to help me calibrate how long the equivalent assignment should be next year.

Your report should probably include at least the following information:

- What are the key engineering aspects to the solution?
- What types of problems is this solution well suited at addressing?
- What are some challenges or downsides of the solution?
- What situations would the solution be more or less appropriate?
- Are there any particular cultural considerations for this type of project?
- How will the solution be impacted by climate change?
- Any other important questions that relate to your specific engineering intervention.